1) Find out the list of Companies which will ask DSA for the Placement drive

Amazon
microsoft
Google
Adobe
Samsung
Intel
Walmart Global Tech
Oracle
NVIDIA
Pay Pal
American express

2) Find out the gate syllabus for DSA

Unit 1 : Programming and Data structures

1. Introduction to Data structures
2. Arrays
3. Linked Lists
4. Stacks
5. Queues
6. Hash Tables (Hashing)
7. Trees
8. Graphs
9. Recursion

Unit2 : Algorithms

1. Asymptotic Analysis
2. Searching Algorithms
3. Sorting Algorithms
4. Hashing
5. Divide and Conquer
6. Greedy Algorithms

7. Dynamic programming
8. Graph Algorithm

3) Find set of inputs and outputs for following

1. electricity bill

Inputs :
Consumer Number
Consumer Name
Previous Meter Reading
Current Meter Reading
Tariff Type
Billing Month

Outputs :
Units Consumed
Energy charges
Fixed charges
Taxes (if any)
Total Bill Amount
Due Date
Bill Status

~~Outputs~~ :

2. Bus ticket

Inputs :
Passenger Name
Source
Destination
Travel Date
Number of Passengers
Bus Type
Seat Numbers

Outputs :
Ticket Number
Bus Number
Departure Time
Arrival Time
Seat Numbers
Fare
Booking status (confirmed/waiting)
Boarding Point

3. Message to a friend

Inputs	Outputs
Sender Name	Message Displayed
Friend's Name / Contact	Sent status (sent / delivered)
Message Text	Delivery Time
Date and Time	Read status (seen / unread)

Today when you started from the home to college, what are the various data structures identified in your surroundings.

Data structures	Real-life example
Array	Row of Parked bikes and Cars
Queue	lines of people at bus stop
Stack	stack of books
linked list	Bus route map
2D array	Classroom seating arrangement
Priority Queue	Ambulance in traffic
Circular Queue	vehicles moving around a traffic roundabout